

K-S Test

Question

- Consider 5 Random numbers 0.723, 0.234, 0.655, 0.435, 0.117 test for uniformity using K-S Testing for significance level 5 %

Form Question:

$N = 5$ (no of random numbers)

$\alpha = 0.05$ (Significance level 5 %)

$R_i = \{0.117, 0.234, 0.435, 0.655, 0.723\}$ (Random numbers sorted in ascending order)

Step 1 : Tablular form

i	1	2	3	4	5
Ri	0.117	0.234	0.435	0.655	0.723
i/N	0.2	0.4	0.6	0.8	1
(i/N)- Ri	0.083	0.166	0.165	0.145	0.277
(i-1)/N	0	0.2	0.4	0.6	0.8
Ri-{(i-1)/N}	0.117	0.034	0.035	0.055	

Identify D^+ , D^- , D

$$D^+ = \max\{(i/N) - R_i\} = 0.277 \text{ (max in row } (i/N) - R_i)$$

$$D^- = \max\{R_i - ((i-1)/N)\} = 0.117 \text{ (max in row } R_i - ((i-1)/N))$$

$$D = \max\{D^+, D^-\} = \max\{0.277, 0.117\} = 0.277$$

Sample Size, N	Level of Significance, α				
	0.20	0.15	0.10	0.05	0.01
3	0.565	0.597	0.642	0.708	0.828
4	0.494	0.525	0.564	0.624	0.733
5	0.446	0.474	0.474	0.565	0.669
10	0.322	0.342	0.368	0.410	0.490
15	0.266	0.283	0.304	0.338	0.404
20	0.231	0.246	0.264	0.294	0.356
25	0.21	0.22	0.24	0.27	0.32
30	0.19	0.20	0.22	0.24	0.29
35	0.18	0.19	0.21	0.23	0.27
40	0.17	0.18	0.19	0.21	0.25
45	0.16	0.17	0.18	0.20	0.24
50	0.15	0.16	0.17	0.19	0.23
over] 50]	$\frac{1.07}{\sqrt{N}}$	$\frac{1.14}{\sqrt{N}}$	$\frac{1.22}{\sqrt{N}}$	$\frac{1.36}{\sqrt{N}}$	$\frac{1.63}{\sqrt{N}}$

Identify D_α form N and α

Here,

$$N = 5$$

$$\alpha = 0.05$$

$$D_\alpha = 0.565$$

Condition Checking

- $D < D_{\alpha}$, the sequence of random no. is uniformly distributed & the null hypothesis is not rejected.
- Here,
- $D = 0.277$
- $D_{\alpha} = 0.565$
- $D < D_{\alpha}$; the sequence of random no. is uniformly distributed & the null hypothesis is not rejected.